

MLFF-DATA9A6c Observable Evidence and Leakage Closure Specification

mdstats project

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1 Purpose

DATA9A6c closes the remaining integrity defects in the analysis-owned observable bridge before general material-profile migration. It changes evidence identity and dependency ordering but does not change any RDF, coordination, dynamics, transport, or thermomechanical algorithm.

2 Collection identity verification

Supplied collection identities are verification inputs, never trusted substitutes for identities recomputed from the analyzed arrays.

The bridge **MUST** recompute an **ObservableCollectionIdentity** from every collection actually passed to analysis. A caller-supplied identity **MUST** match the recomputed scientific content digest and expected label or execution fails.

Scientific identity **MUST** be relocation invariant. Filesystem paths are retained only as location hints. Source content digests, when available, remain identity fields. Object-dtype arrays **MUST** be rejected.

Composition evidence **MUST** include species counts and an atomic-number sequence digest. The explicit sequence may be omitted for large systems.

3 Symmetric generation lineage

Reference and candidate trajectories MUST use one `TrajectoryGenerationIdentity` schema. The record MUST include generator kind, artifact and protocol digests, output collection digest, engine/version, and precision policy. Optional fields include manifest, runtime environment, initial configuration, source provenance, seed, and notes.

With `require_complete_lineage=True`, both records are mandatory and each `output_collection_digest` MUST equal the analyzed collection identity. The legacy `MLFFTrajectoryGenerationIdentity` is a compatibility wrapper and MUST provide an explicit output digest in complete-lineage mode.

4 Analysis-owned result identity

Every registered observable result MUST produce an `ObservableResultIdentity` containing call ID, observable ID, native result type, serializer ID, and canonical SHA-256 digest. The analysis facade owns this identity. It MUST support current dataclass, mapping, sequence, enum, scalar, and NumPy-array result structures and reject unstable opaque/object-dtype representations.

The MLFF bridge MUST reference these result identities. It MUST NOT duplicate or redefine scientific arrays.

5 Statistical role and leakage control

Every plan MUST declare one `ObservableEvidenceRole`:

```
training_diagnostic
checkpoint_monitor
outer_validation
calibration
locked_test
external_benchmark
```

An `ObservableValidationActivationRecord` MUST bind the upstream identities required by the role. Outer validation, calibration, and locked test require partition policy and assignment. Outer validation, calibration, locked test, and external benchmark require a predeclared comparison-policy digest. Locked test additionally requires protocol-freeze and evaluation-activation digests.

The dependency direction is normative:

```
comparison policy + activation -> evidence -> comparison result -> decision
```

The following reverse dependencies are forbidden:

- realized evidence to comparison-policy fitting;
- locked-test evidence to feature fitting, selection, protocol choice, checkpoint selection, calibration policy, or acquisition;
- post-training physical validation to retroactive dataset selection.

6 Runtime and capability identity

Runtime evidence MUST distinguish executing-source version from installed package metadata. It records source/install mode, executing module path and hash, Python/platform identity, and numerical library versions.

Capability identity MUST include owner source implementation and function signature, stable owner-manual ID, source-path hint, versioned manual URI, parameter codec, and result type.

7 Restorable evidence

`MLFFObservableValidationEvidenceRecord` MUST be JSON-safe, restore without loading native result arrays, and reject any content-digest mismatch.

8 Documentation and packaging

Automated **source/wheel registry parity** MUST be enforced: the source package and wheel MUST expose the same capability registry. The wheel MUST include the owner-manual index. **Valid JSON release artifacts** MUST contain JSON only; console warnings belong in separate stderr/log files. Distributed checksums MUST use relative artifact names.

9 Acceptance tests

DATA9A6c passes only when tests cover:

1. supplied identity mismatch rejection;
2. relocation-invariant identity;
3. object-dtype rejection;
4. symmetric reference/candidate generation lineage;
5. output-collection mismatch rejection;
6. locked-test activation gates;
7. result identities for all 22 registered calls;
8. evidence-record round trip and tamper rejection;
9. source/wheel registry parity and packaged manual index;
10. dependency graph acyclicity and forbidden leakage edges;
11. valid JSON release artifacts;
12. updated manuals and rendered PDFs.